



POLITECHNIKA
LUBELSKA
WYDZIAŁ ELEKTROTECHNIKI
I INFORMATYKI

Programowanie Aplikacji w Chmurze Obliczeniowej

Laboratorium 5

WIELOETAPOWE BUDOWANIE OBRAZÓW

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Treść Dockerfile

```
FROM scratch AS builder

ARG VERSION=1.0.0

ADD alpine-minirootfs-3.21.3-aarch64.tar /

RUN apk add --no-cache nodejs npm go

WORKDIR /app

COPY index.js .

RUN echo "<html><body> \
<h1>Informacje o serwerze</h1> \
<p><b>Wersja aplikacji:</b> ${VERSION}</p> \
<p><b>Hostname:</b> $(hostname)</p> \
<p><b>Data buildu:</b> $(date)</p> \
</body></html>" > index.html

FROM nginx:alpine

COPY --from=builder /app/index.html /usr/share/nginx/html/index.html
COPY --from=builder /app/index.js /usr/share/nginx/html/index.js

HEALTHCHECK --interval=10s --timeout=3s \
  CMD wget -qO- http://localhost/ || exit 1

EXPOSE 80
```

Treść index.js

```
const express = require('express');
const os = require('os');

const app = express();
const PORT = 8080;

const VERSION = process.env.APP_VERSION || 'unknown';

app.get('/', (req, res) => {
  const ipAddresses = Object.values(os.networkInterfaces())
    .flat()
    .filter(details => details.family === 'IPv4' &&
!details.internal)
    .map(details => details.address);

  const hostname = os.hostname();

  res.json({
    server_ip: ipAddresses[0],
    hostname: hostname,
    version: VERSION
  });
});

app.listen(PORT, () => {
  console.log(`Server is running on port ${PORT}`);
  console.log(`Version: ${VERSION}`);
});
```

Polecenia docker

Budowanie obrazu:

```
docker build --build-arg VERSION=0.0.1 -t app1:v0.0.1 .
```

Uruchamianie kontenera:

```
docker run -d -p 8080:80 --name app app1:v0.0.1
```

```
PS C:\Users\Home\lab5> docker run -d -p 8080:80 --name app app1:v0.0.1 6d178bbd42de4a2ad412fbcbcac721501d4c3f6325fe38b02874b993772ac54f
```

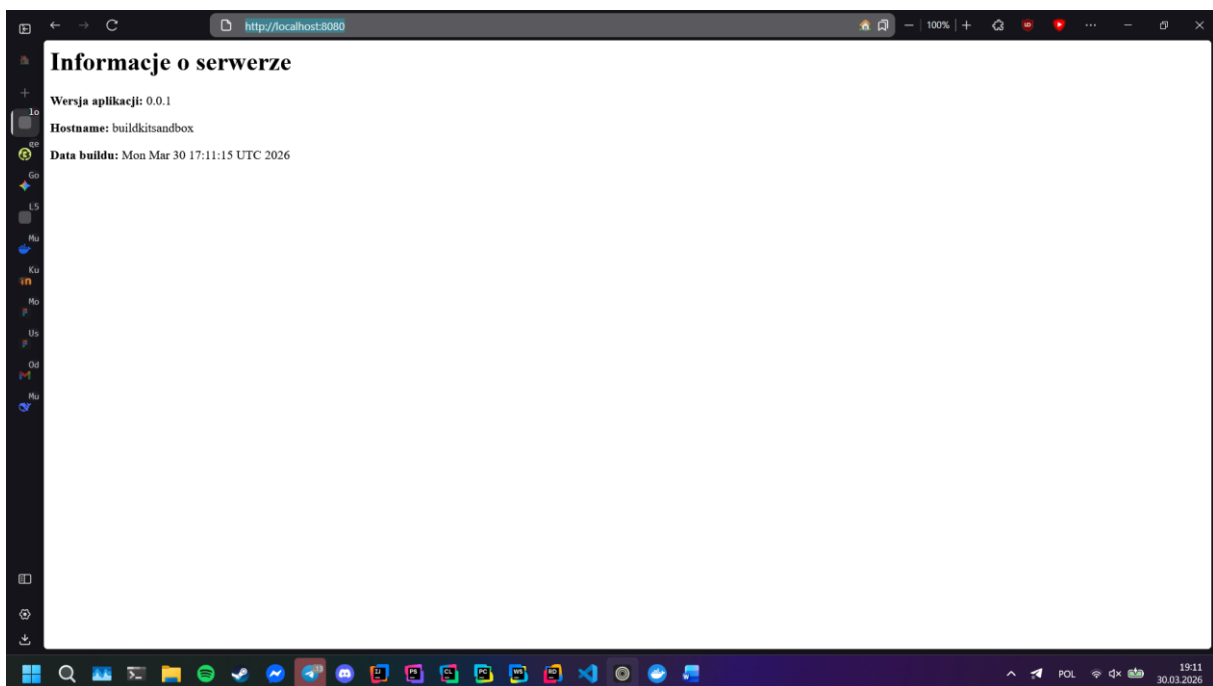
Polecenie uruchamiające i wynik

		app	6d178bbd42de	app1:v0.0.1	8080:80	0%	20 minutes ago				
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Uruchomiony kontener

```
PS C:\Users\Home\lab5> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
6d178bbd42de   app1:v0.0.1   "/docker-entrypoint..." 20 minutes ago Up 20 minutes (healthy) 0.0.0.0:8080->80/tcp, [
::]:8080->80/tcp   app
PS C:\Users\Home\lab5> \
```

Uruchomiony kontener



Działająca strona